MANIPAL INSTITUTE OF TECHNOLOGY
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MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL

DEPARTMENT OF MATHEMATICS

III SEMESTR B.TECH. (AI&ML ENGINEERING)

END SEMESTER EXAMINATION

Subject : LINEAR ALGEBRA AND LOGIC MAT 2128

Date of Examination: 02/12/2024 Time: 9.30 AM to 12.30 PM MAX.MARKS: 50

Q. No	Question	M	CO	PO	Blooms Taxonomy level
1A	Find a generating set for null space of $A = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 2 & 4 & 0 & -8 \\ 0 & 0 & 2 & 6 \end{bmatrix}$. Check whether the vector $u = \begin{bmatrix} 2 \\ -3 \\ 3 \\ -1 \end{bmatrix}$ is in null space of A?	4	1		1,2,3
1B	Let $y = \begin{bmatrix} 3 \\ 1 \\ 5 \\ 1 \end{bmatrix}, u_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$ and $W = Span\{u_1, u_2\}$. Find the closest point to y in the subspace W spanned by u_1 and u_2.	3	2		2,3
1C	Obtain the principal disjunctive normal form and principal conjunctive normal form of $f(x_1, x_2, x_3) = \overline{(x_1 \vee x_2)} \vee (\overline{x_1} \wedge x_3)$.	3	5		2,3

2A	Let $T: R^2 \to R^3$ be a linear transformation defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 3x_1 - x_2 \\ -x_1 + 2x_2 \\ 2x_1 \end{bmatrix}$. a) Determine a generating set for range of T b) Determine a generating set for null space of T c) Is T onto? d) Is T a one to one mapping?	**4**	**1**		2,3
2B	Give Cholesky's decomposition of $A = \begin{pmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{pmatrix}$	**3**	**3**		1,2,3
2C	Show that $R \wedge (P \vee Q)$ is a valid conclusion from the premises $P \vee Q, Q \to R, P \to M,$ and ꟾM.	**3**	**5**		1,2,3
3A	Find singular value decomposition of $A = \begin{pmatrix} 4 & 0 \\ 3 & -5 \end{pmatrix}$	**4**	**4**		2,3
3B	For what value(s) of `h' will y be in the subspace of R^3 spanned by v_1, v_2, v_3 if $v_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix}, v_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$ and $y = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$.	**3**	**1**		2,3
3C	Diagonalize the matrix $A = \begin{pmatrix} 4 & -3 & 0 \\ 2 & -1 & 0 \\ 1 & -1 & 1 \end{pmatrix}$ if possible.	**3**	**4**		2,3
4A	Use eigenvalues and eigenvectors to solve the initial value problem $y_1' = 4y_1 + y_2$ $y_2' = 3y_1 + 2y_2$ Subjected to $y_1(0) = 120$ and $y_2(0) = 40$	**4**	**3**		2,3
4B	Let $v_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 7 \\ 1 \end{bmatrix}$ and $p = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$. Write p as an affine combination of v_1, v_2 and v_3 if possible	**3**	**1**		2,3

4C	Show that (∃x) M(x) follows logically from the premises $(x)(H(x) \rightarrow M(x))$ and (∃x) H(x)	**3**	**5**		2,3
5A	Solve the system by QR decomposition method. $x_1 + 2x_2 + x_3 = 1$ $x_1 + x_2 + x_3 = 3$ $x_1 + 2x_3 = 6$	**4**	**2**		2,3
5B	For all $u, w \in V$ where V is an inner product space, prove that $\lvert< u, w >\rvert \leq \lVert u\rVert\lVert w\rVert$	**3**	**2**		1,2,3
5C	Make the change of variable that transforms the quadratic form $Q(x) = x_1^2 - 6x_1x_2 + 9x_2^2$ in to a new quadratic form which has no cross-product terms.	**3**	**4**		2,3